App Development Project Report

1. Student Details

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About Me: I am currently a second-year CSE undergraduate and a student in the IIT Madras BS Degree program. I have a keen interest in web development as well as emerging fields like machine learning, artificial intelligence, and data science. My passion for data structures and algorithms fuels my drive to build meaningful, data-driven applications that combine solid analytics and engaging user experiences. I enjoy leveraging my growing skill set to tackle real-world challenges and expand my understanding in the rapidly evolving tech landscape.

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2. Project Details

Project Title:Hospital-Management System(HealthCare+)

Problem Statement:

Flask based application for managing hospitals with CRUD and database

Approach:

The project was developed using Flask as the backend framework with a modular, extensible codebase. Users can manage and update data seamlessly through intuitive interfaces, while the system supports features like appointment scheduling, profile management, and real-time status tracking. Key interactions and patterns are visualized using interactive charts, enabling users to monitor trends and extract actionable insights over time.

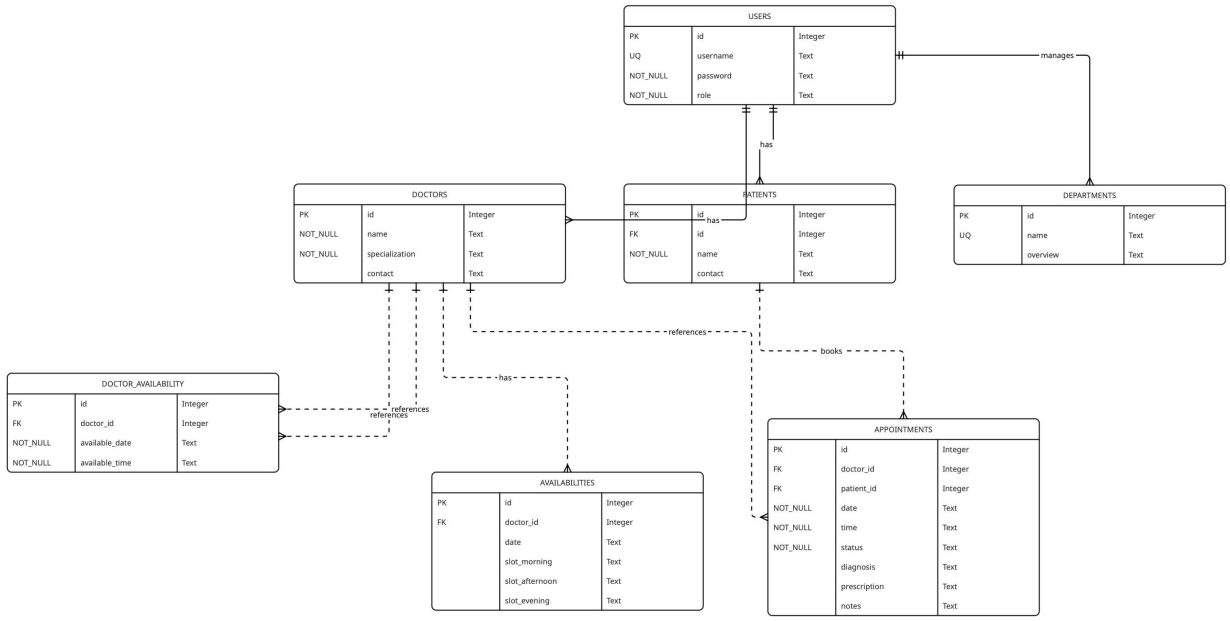
3. AI/LLM Declaration

I employed ChatGPT (GPT-4/5) to assist with front-end styling suggestions, optimizing JavaScript logic for interactivity, and generating charting code samples for data visualization components. The overall extent of AI/LLM involvement was approximately 10–15%, exclusively limited to code refinement and UI enhancement tasks. All functional integration, backend development, debugging, and final validation were conducted manually by me to ensure code integrity and original implementation.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Back-end web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Chart.js	To visualize trends such as activity vs. time
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing user data

5. Database Schema / ER Diagram



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6. Architecture and Features

Architecture Overview:

- **app.py** – Main Flask application file managing routing, authentication, and all business logic.
- **/models** – database models using SQLAlchemy
- **/routes** – Flask Blueprints for user and activity routes
- **/templates** – Directory of Jinja2 HTML templates for rendering user-facing pages and dashboards.
- **/static** – Contains project CSS, JavaScript (including chart logic), and static images/assets.
- **hospital.db** – SQLite database file for storing all application data.
- **schema.sql** – File with complete database schema for initialization and migrations.
- **README.md** – Instructions and overview for running and understanding the project.

Implemented Features:

User registration and login
Daily activity tracking and habit logging
Visualization dashboard using Chart.js
Summary analytics with date filtering
Responsive UI built with Bootstrap
Secure password hashing and user session handling

Additional Features:

- Export the appointments, doctors list and patients list as CSV for easier accessibility.
- Integrated profile photo upload and live update for doctors and patients, enhancing personalization.

7. Video Presentation

Drive Link:

https://drive.google.com/file/d/1iaqx8nN6Mqm2fDJaXps_NdOJaMfYg_Q6/view?usp=sharing